TRIGONOMETRY

(त्रिकोणमिति)

SHEET-04

CLASS ROOM SHEETS / FOR ALL EXAMS

UPDATED

TRIGONOMETRY (त्रिकोणमिति)

CLASSROOM SHEET-04

1. The value of $(1+\sin^4A-\cos^4A)\operatorname{cosec}^2A$ is:

SSC CPO 03/10/2023 (Shift-02)

(a) −1 (b) 1
(c) −2 (d) 2

2. $(\cot\theta+\tan\theta)(\operatorname{cosec}\theta-\sin\theta)(\cos\theta-\sec\theta)$ = ----.

SSC CHSL 03/07/2024 (Shift-01)

(a) 0 (b) 1
(c) 2 (d) −1

3. Find the value of
$(\operatorname{cosec}\theta-\sin\theta)(\sec\theta-\cos\theta)(\tan\theta+\cot\theta)$.

SSC CGL 25/09/2024 (Shift-01)

(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) −1

4. The value of

$$\left(\frac{\sin A}{1-\cos A}+\frac{1-\cos A}{\sin A}\right)\div\left(\frac{\cot^2A}{1+\operatorname{cosec}A}+1\right)$$ is:

SSC CGL TIER-II 12/09/2019

(a) $\frac{3}{2}$ (b) $\frac{1}{2}$
(c) 1 (d) 2

5. The value of

$$\frac{\sin A}{\cot A+\operatorname{cosec}A}-\frac{\sin A}{\cot A-\operatorname{cosec}A}-1$$

SSC CPO 11/11/2022 (Shift-02)

(a) $\frac{1}{2}$ (b) 1
(c) 0 (d) 2

6. The expression $(\cos^6\theta+\sin^6\theta-1)(\tan^2\theta+\cot^2\theta+2)+3$ is equal to:

SSC CGL 13/04/2022 (Shift-02)

(a) 0 (b) − 1
(c) 2 (d) 1

7. Let $0°<\theta<90°$, $(1+\cot^2\theta)(1+\tan^2\theta)\times(\sin\theta-\operatorname{cosec}\theta)(\cos\theta-\sec\theta)$ is equal to:

SSC CGL TIER-II 29/01/2022

(a) $\sec\theta\operatorname{cosec}\theta$ (b) $\sec\theta+\operatorname{cosec}\theta$
(c) $\sin\theta+\cos\theta$ (d) $\sin\theta\cos\theta$

8. The expression

$$\frac{\tan^6\theta-\sec^6\theta+3\sec^2\theta\tan^2\theta}{\tan^2\theta+\cot^2\theta+2}$$ $0°<\theta<90°$, is equal to:

SSC CGL TIER-II 03/02/2022

(a) $\sec^2\theta\operatorname{cosec}^2\theta$ (b) $-\sec^2\theta\operatorname{cosec}^2\theta$
(c) $\cos^2\theta\sin^2\theta$ (d) $-\cos^2\theta\sin^2\theta$

9. The expression

$$\frac{(1-\sin\theta+\cos\theta)^2(1-\cos\theta)\sec^3\theta\operatorname{cosec}^2\theta}{(\sec\theta-\tan\theta)(\tan\theta+\cot\theta)}$$, $0°<\theta<90°$, is equal to

SSC CGL TIER-II 03/02/2022

(a) $\sin\theta$ (b) $2\cos\theta$
(c) $\cot\theta$ (d) $2\tan\theta$

10. The expression

$$\frac{(1-2\sin^2\theta\cos^2\theta)(\cot\theta+1)\cos\theta}{(\sin^4\theta+\cos^4\theta)(1+\tan\theta)\operatorname{cosec}\theta}-1$$, $0°<\theta<90°$, is equal to:

SSC CGL TIER-II 03/02/2022

(a) $\cos^2\theta$ (b) $-\sin^2\theta$
(c) $\sec^2\theta$ (d) $-\sec^2\theta$

11. If $\cot\alpha=\frac{P}{Q}$, then find the value of

$$\frac{P\cos\alpha-Q\sin\alpha}{P\cos\alpha+Q\sin\alpha}-\frac{P^2-Q^2}{P^2+Q^2}+3.$$

SSC CHSL 09/07/2024 (Shift-01)

(a) $\frac{P_2}{P_2+Q_2}$ (b) $\frac{P}{Q}$
(c) 0 (d) 3

12. If m = asecA and y = btanA, then find the value of
$b^2m^2-a^2y^2+\frac{a^2y^2}{b^2m^2}+\cos^2A$.

SSC CGL 13/09/2024 (Shift-03)

(a) a^2b^2 (b) $1-a^2b^2$
(c) a^2b^2+2 (d) a^2b^2+1

13. If m = secθ − tanθ and n = cosecθ + cotθ, then what is the value of m + n(m − 1)?

SSC CHSL 13/03/2023 (Shift-02)

(a) 2 (b) 1
(c) 0 (d) −1

14. Which of the following is true when x = sinA + cosA; y = secA + cosecA?

SSC CPO 27/06/2024 (Shift-03)

(a) $y(1+x^2)=2x$ (b) $y-2x=x^2y$
(c) $y+2x=x^2y$ (d) $y(1-2x^2)=x$

15. If cotA + cosA = p and cot A − cosA = q. then which of the following relation is correct.?

SSC CHSL 03/08/2023 (Shift-04)

(a) $\sqrt{\frac{1}{16pq}}=p^2+q^2$ (b) $\frac{1}{4pq}=p^2+q^2$
(c) $\sqrt{16pq}=p^2-q^2$ (d) $4pq=p^2+q^2$

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16. If tanA + sinA = x, tanA – sinA = y, what is the value of $x^2 - y^2$?

SSC CHSL 11/07/2024 (Shift-04)

(a) $2\sqrt{\frac{x}{y}}$ (b) $4\sqrt{\frac{x}{y}}$

(c) $\sqrt{xy}$ (d) $4\sqrt{xy}$

17. If (sin x + sin y) = a and (cos x + cos y) = b, then find the value of (sin x sin y + cos$x$ cos y).

SSC CPO 27/06/2024 (Shift-02)

(a) $\frac{a^2+b^2-1}{2}$ (b) $\frac{a^2+b^2-1}{2}$

(c) $\frac{a^2+b^2-2}{2}$ (d) $\frac{a^2+b^2+2}{2}$

18. If asin³ X + bcos³ X = sinXcosX and asinX = bcosX, then find the value of $a^2 + b^2$, provided that X is neither 0° nor 90°.

SSC CGL 11/09/2024 (Shift-01)

(a) 0 (b) 1

(c) $a^2 - b^2$ (d) $a^2 + b^2$

19. [(sinθ tanθ + cosθ)² – 1] is equal to:

SSC Phase-X 02/08/2022 (Shift-02)

(a) $\sec^2\theta$ (b) $\sec\theta$

(c) $\text{cosec}\,\theta$ (d) $\tan^2\theta$

20. The value of

$\left(\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}\right)^3 \frac{1}{\tan\theta} - \sec^2\theta$ is:

SSC CGL 12/09/2024 (Shift-01)

(a) 2 (b) 1

(c) 0 (d) –1

21. $\frac{\cot^3\theta}{\text{cosec}^2\theta} + \frac{\tan^3\theta}{\sec^2\theta} + 2\sin\theta\cos\theta = ?$

SSC CGL 16/08/2021 (Shift-02)

(a) $\text{cosec}\theta\sec\theta$ (b) $\text{cosec}^2\theta\sec^2\theta$

(c) $\sin\theta\cos\theta$ (d) $\sin\theta$

22. $\frac{1+\cos\theta-\sin^2\theta}{\sin\theta(1+\cos\theta)} \times \frac{\sqrt{\sec^2\theta+\text{cosec}^2\theta}}{\tan\theta+\cot\theta}$, $0° < \theta < 90°$

SSC CGL TIER-II 29/01/2022

(a) $\tan\theta$ (b) $\sec\theta$

(c) $\text{cosec}\theta$ (d) $\cot\theta$

23. The expression

$\frac{\cos^4\theta - \sin^4\theta + 2\sin^2\theta + 3}{(\text{cosec}\,\theta + \cot\theta + 1)(\text{cosec}\,\theta - \cot\theta + 1) - 2}$ $0° < \theta < 90°$, is equal to

SSC CGL TIER-II 03/02/2022

(a) $\frac{1}{2}\sin\theta$ (b) $2\sin\theta$

(c) $\sec\theta$ (d) $2\text{cosec}\theta$

24. Using trigonometric formulas, find the value of

$\left(\frac{\sin(x-y)}{\sin(x+y)}\right)\left(\frac{\tan x + \tan y}{\tan x - \tan y}\right)$

SSC CGL 18/07/2023 (Shift-01)

(a) –2 (b) 2

(c) 0 (d) 1

25. If tanA = a tanB and sinA = b sinB, then the value of cos²A is:

SSC CHSL 05/07/2024 (Shift-01)

(a) $\frac{b^2-1}{a^2-1}$ (b) $\frac{a^2-1}{b^2-1}$

(c) $\frac{b^2+1}{a^2+1}$ (d) $\frac{a^2+1}{b^2+1}$

26. The value of the expression

$\frac{1-\sin(2t)}{1+\sin(2t)} \times \frac{\cos(t)+\sin(t)}{\cos(t)-\sin(t)}$ is:

SSC CGL 19/09/2024 (Shift-02)

(a) $\frac{1-2\tan(t)}{1+2\tan(t)}$ (b) $\frac{1-\tan(t)}{1+\tan(t)}$

(c) $\frac{1+2\tan(t)}{1-2\tan(t)}$ (d) $\frac{1+\tan(t)}{1-\tan(t)}$

27. Using $\text{cosec}(\alpha+\beta)$

$= \frac{\sec\alpha\times\sec\beta\times\text{cosec}\alpha\times\text{cosec}\beta}{\sec\alpha\times\text{cosec}\beta+\text{cosec}\alpha\times\sec\beta}$,

find the value of cosec 75°.

SSC CGL 25/07/2023 (Shift-01)

(a) $1\frac{\sqrt{6}+\sqrt{2}}{4}$ (b) $\frac{\sqrt{6}-\sqrt{2}}{4}$

(c) $\sqrt{6}-\sqrt{2}$ (d) $\sqrt{6}+\sqrt{2}$

28. The value of the expression $4(\sin^6A + \cos^6A) - 6(\sin^4A + \cos^4A) + 8$ is:

SSC CHSL 02/08/2023 (Shift-04)

(a) 4 (b) 8

(c) 7 (d) 6

29. Find the value of the following expression. $12(\sin^4\theta + \cos^4\theta) + 18(\sin^6\theta + \cos^6\theta) + 78\sin^2\theta\cos^2\theta$

SSC CGL 18/07/2023 (Shift-03)

(a) 30 (b) 40

(c) 10 (d) 20

30. $\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$ is equal to :

SSC CGL TIER-II 11/09/2019

(a) 1 + sinA – cosA (b) 1 – sinA.cosA

(c) 1 + cosA – sinA (d) 1 + sinA.cosA

31. Find the value of the following expression $5(\sin^4\theta + \cos^4\theta) + 3(\sin^6\theta + \cos^6\theta) + 19\sin^2\theta\cos^2\theta$.

SSC CGL 07/12/2022 (Shift-03)

(a) 8 (b) 5

(c) 6 (d) 7

32. The value of

$\frac{\sec^6\theta - \tan^6\theta - 3\sec^2\theta\tan^2\theta + 1}{\cos^4\theta - \sin^4\theta + 2\sin^2\theta + 2}$ is:

SSC CGL 05/03/2020 (Shift-01)

(a) $\frac{2}{3}$ (b) 1

(c) $\frac{3}{4}$ (d) $\frac{1}{2}$

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33. What is the value of $\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$?

SSC CPO 29/06/2024 (Shift-03)

(a) 0 **(b)** 1

(c) −1 **(d)** 2

34. If $\cot^2\theta = 1 - e^2$, then the value of $\text{cosec}\,\theta + \cot^3\theta \sec\theta$ is:

SSC CPO 05/10/2023 (Shift-03)

(a) $(2 - e^2)^{\frac{3}{2}}$ **(b)** $(1 - e^2)$

(c) $(1 - e^2)^{\frac{3}{2}}$ **(d)** $(2 - e^2)^{\frac{1}{2}}$

35. If $\tan^2\theta = 1 - a^2$, then the value of $\sec\theta + \tan^3\theta\,\text{cosec}\theta$ is:

SSC CGL 01/12/2022 (Shift-04)

(a) $(2 - a)^{\frac{3}{2}}$ **(b)** $(a^2 - 1)^{\frac{3}{2}}$

(c) $(2 - a^2)^{\frac{3}{2}}$ **(d)** $a^{\frac{3}{2}}$

36. If $\tan^2\alpha = 3 + Q^2$, then $\sec\alpha + \tan^3\alpha\,\text{cosec}\alpha$ = ?

SSC CGL 07/12/2022 (Shift-01)

(a) $(3 + Q^2)^{\frac{3}{2}}$ **(b)** $(7 + Q^2)^{\frac{3}{2}}$

(c) $(5 - Q^2)^{\frac{3}{2}}$ **(d)** $(4 + Q^2)^{\frac{3}{2}}$

ANSWER KEY

1.	(d)	2.	(d)	3.	(c)	4.	(d)	5.	(b)	6.	(a)	7.	(a)	8.	(d)	9.	(d)	10.	(b)
11.	(d)	12.	(d)	13.	(d)	14.	(c)	15.	(c)	16.	(d)	17.	(c)	18.	(b)	19.	(d)	20.	(d)
21.	(a)	22.	(d)	23.	(b)	24.	(d)	25.	(a)	26.	(b)	27.	(c)	28.	(d)	29.	(a)	30.	(a)
31.	(a)	32.	(a)	33.	(c)	34.	(a)	35.	(c)	36.	(d)								

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CALCULATION TECHNIQUES

TECHNIQUE - 01

♦ Cube Root of Non-perfect cube ♦

$$\sqrt[3]{N} = A \pm \frac{B}{3.A^2}$$

First find a number which is a perfect cube near to the number N, let it be A and the difference between N and A^3 is B.

Note: '+' sign is used if $N > A^3$ & '–' sign is used if $N < A^3$

Ex. $\sqrt[3]{60} = 4 - \frac{4}{3 \times 4^2} = \frac{188}{48} = 3.92$

Ex. $\sqrt[3]{72} = 4 + \frac{8}{3 \times 4^2} = \frac{200}{48} = 4.17$

♦ Do it Yourself ♦

(i) $\sqrt[3]{10}$ = ...

(ii) $\sqrt[3]{66}$ = ...

(iii) $\sqrt[3]{118}$ = ...

(iv) $\sqrt[3]{226}$ = ...

(v) $\sqrt[3]{525}$ = ...

(vi) $\sqrt[3]{1365}$ = ...

(vii) $\sqrt[3]{1500}$ = ...

TECHNIQUE - 02

♦ Square of two digit number ♦

+ Carry + Carry

$(ab)^2 = a^2/2ab/b^2$

1 digit, 1 digit

- **Square of 14**

Here, a = 1 & b = 4

$14^2 = 1^2/2.1.4/4^2 = 1/8/16 = 196$

- **Square of 23**

Here, a = 2 & b = 3

$23^2 = 2^2/2.2.3/3^2 = 4/12/9 = 529$

♦ Do it Yourself ♦

(i) 18^2 = ... (ii) 23^2 = ...

(iii) 27^2 = ... (iv) 34^2 = ...

(v) 38^2 = ... (vi) 43^2 = ...

(vii) 57^2 = ... (viii) 64^2 = ...

(ix) 66^2 = ... (x) 75^2 = ...

(xi) 78^2 = ... (xii) 84^2 = ...

SELF MENTORSHIP TABLE

	STEP-01	STEP-02	STEP-03	STEP-04	STEP-05	STEP-06
TOPIC NAME	VIDEOS	CLASSROOM SHEET	SSC MATHS	NOTES / FORMULA BOOK REVISION	CHAPTER TEST	PRO LEVEL PRACTICE SHEET
TRIGONOMETRY SHEET - 04						
CONCEPT OF VALUE PUTTING						